



Three-dimensional tunnel face extrusion and reinforcement effects of underground excavations in deep rock masses

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ABSTRACT

The design experience and the shallow tunnel theory, which emphasize the plane strain assumption, two-dimensional (2D) strength theory, and indoor or back analysis-based parameters of rock mass, should not be directly adopted for deep tunnel design. This paper reports the development of a digital acquisition technique to quickly, automatically, and accurately obtain the in-situ strength parameters (e.g., GSI and D), followed by the establishment of a three-dimensional (3D) numerical model for deep tunneling for a typical tunnel in western China to study the 3D and nonlinear spatial effects and mechanisms during excavation with incorporation of the 3D Hoek-Brown (HB) strength criterion. The reliability and correctness of the developed methods were verified by the field monitoring data, and the results of our study showed that: (1) The deformations based on 2D strength criteria (e.g., HB and MC) and 3D criterion (e.g. GZZ) are close for shallow tunnel, but significantly different for deep tunnel because of the widely distributed plastic zone in deep tunneling, thereby indicating the obvious effect of intermediate principal stress (σ_2) on rock deformation; (2) The face extrusion deformation exhibits obvious nonlinear trends with an increasing buried depth, and the σ_3 of the core rock reflects a tensile value in deep tunnel, thereby indicating the presence of an extremely unstable state. Reinforcement could increase the overall stress inside the core rock and reduce the stress unloading effect caused by the strong disturbances of deep excavations, thereby promoting a gradual change from tensile stress to compressive stress; (3) A uniform relationship was found between pre-extrusion and pre-convergence deformations, indicating that the pre-convergence deformation of the surrounding rock could be controlled by constraining the pre-extrusion deformation of the core rock. This digital technique and such 3D HB-based (e.g., GZZ) numerical modeling may be promising for conducting the 3D forward and dynamic analyses of tunnel stability. Additionally, the core rock extrusion control can be treated as an active method to render the rock itself a natural support material and to realize its strength potential, thereby making a valuable contribution to the construction of deep and ultra-deep rock tunnels.

1. Introduction

The intelligent construction of deep tunnels is of particular interest in the context of current tunnel engineering, wherein the refined collection of rock mass parameters in addition to the precise analysis and control of the tunnel stability are the core contents of intelligent construction. However, it remains a challenge to fully realize tunnel intelligence because the traditional model does not accurately reflect the mechanical behavior of an in-situ rock mass (e.g., linear or two-dimensional (2D) strength theories), the parameters for numerical calculations are still determined by indoor experiments or by back analysis, and three-

dimensional (3D) numerical modeling cannot truly simulate the practical tunneling process. These difficulties are amplified in the context of deep tunneling because of the obvious 3D stress state, the wide distribution of the plastic zone, and the significant 3D space effect near the excavation face. And there is still a lack of an effective 3D forward analysis method that can be used in deep rock tunneling.

The mechanical behavior of deep rock engineering under high ground stresses and strong geo-mechanical disturbances has obvious nonlinear characteristics, which are significantly different from those of shallow rock engineering.^{1–7} The plane strain model and 2D theory are often used in shallow tunnels,^{8,9} and so it has been considered that the

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deformation and stability of surrounding rock depend only on the radial crown settlement and horizontal convergence displacement around the tunnel. However, it is unreasonable to use this simple model and theory for the design and construction of a deep rock tunnel because the excavation process of a deep tunnel exhibits obvious 3D space effects,¹⁰ especially in terms of the extrusion effects at the tunnel face.^{11,12} Indeed, many researchers have confirmed the existence of extrusion deformation and the 3D space effects of tunnel faces through physical model tests and practical field measurements.^{13–26} Meanwhile, disasters (e.g., rock bursts, large deformations, and steel arch distortion) in deep tunnel operations mostly occur close to the tunnel face, while almost no severe disasters occur in the areas far from the tunnel face where the tunnel can be treated as a plane strain model. These tunnel disasters induced by the spatial effects close to the tunnel face therefore have a significant impact on the surrounding rock stability and construction safety.^{9,18,27–37}

The plane strain tunnel model only considers radial convergence deformation, while a real tunnel includes not only convergence deformation, but also extrusion and pre-convergence deformation of the core surrounding rock. As previously reported, Lunardi¹⁸ believes that controlling the deformation of the tunnel face is key to controlling the overall stability of the surrounding rock, and that the deformation of the surrounding rock starts primarily from the advanced core rock mass in front of the tunnel face, gradually developing backward behind it. The convergent deformation of the surrounding rock is therefore only the last stage of an intricate stress-strain process. In addition, Hoek et al.^{28,38} found that tunnel safety problems caused by extrusion of the tunnel face were not prominent when the convergence strain (i.e., the ratio of the tunnel convergence to its diameter) of the unsupported rock tunnel was $<2.5\%$, thereby indicating that the 2D plane theory is suitable for tunnel design in this case. However, this theory was not applicable when the convergent strain was $>2.5\%$ because of the obvious extrusion effect of the tunnel face. The stability of the tunnel is therefore mainly determined by the 3D space effect, rendering the use of 3D numerical modeling essential for stability analysis.^{11,39} Since deep tunnels under high ground stresses exhibit large deformation characteristics, stability analysis of the surrounding rock should focus on the 3D spatial effects of the tunnel face.

To accurately analyze the 3D space effects and complex stress conditions (i.e., $\sigma_1 > \sigma_2 > \sigma_3$) close to the tunnel face of a deep rock tunnel, the analysis model and theory employed should be based on the reliable 3D and nonlinear theory.⁴⁰ Likewise, the strength theory of deep rock mass should be considered. However, it should be noted that the widely used 2D Mohr-Coulomb (MC) strength criterion and nonlinear Hoek-Brown (HB) criterion cannot accurately reflect the strength characteristics of a deep rock mass under complex stress conditions. Many scholars have therefore carried out related studies and believe that the numerical results obtained using 3D strength criteria are quite different from those obtained using the 2D criteria and the plane strain model.^{9,12,27,29,30,36,41–45} They also argue that reliable results in relation to the tunnel face must rely on the 3D analysis model and theory.³⁹

Previous deep tunnel stability research has focused on the radial convergence deformation of surrounding rock, and there has been little research on the extrusion effects of the tunnel face and the nonlinear law of the extrusion effects for varied tunnel depths. Existing studies on the extrusion effects of the tunnel face have also focused on the deformation and stability of the tunnel face,^{28,46} the influencing factors of extrusion deformation,²⁹ and the reinforcement of the tunnel face based on the MC criterion,^{34,47} the DP criterion,²⁷ and the HB criterion.^{33,42} However, the mechanical mechanism and strength characteristics of tunnel face extrusion, the law between tunnel face pre-extrusion and pre-convergence deformation, and the active control mechanisms based on the tunnel face deformation constraints on the overall stability of the surrounding rock have not been examined to any great extent.

The determination of reliable rock mass strength parameters is therefore essential for the accurate analysis and calculation of the surrounding rock stability. Due to the influence of the scale effect, indoor

parameters and in situ rock mass parameters may differ by orders of magnitude, and any approach that reduces indoor rock parameters to make them applicable to rock mass engineering exhibits inordinate randomness, and so the obtained results cannot meet engineering accuracy requirements. Although rock mass parameters are often back analyzed based on the field deformation response, the parameters obtained are generalized because of the model and the presence of systematic and accidental errors. In addition, the results may be locally optimal and so the real mechanical rock mass processes cannot be considered for multi-parameter back analysis. The back analysis-based design method therefore cannot meet the requirements of dynamic and forward design. Thus, in the intelligent construction of deep tunnels, parameter field acquisition and 3D forward analysis are key technologies. Although many studies have been published in relation to tunnel excavation simulations, the parameter reliability problem has not been resolved, and there remains a lack of a continuous analysis methodology that directly uses field rock mass parameters for 3D forward analysis. It is commonly known that in rock engineering, the mechanical behavior and stability are dominated by complex structural planes such as joints and cracks inside the rock mass, which cannot be accurately determined and recognized by field surveys alone. However, the structural information related to the rock mass, which is reflected by the tunnel face, is the most direct, accurate and valuable information available^{48–51}; despite this, it is often ignored or underutilized.

Considering these limitations, this study employs a digital method, i.e., mobile phone photography or 3D laser scanning combined with 3D reconstruction methods,^{48–50} to obtain in situ geological and rock mass parameters such as the geological strength indicator (GSI) and excavation disturbance coefficient (D) values,^{52,53} while using the 3D nonlinear strength criterion (GZZ) to conduct the 3D forward analysis of deep tunnel excavation. The main focuses of our research include the 3D space effect close to the tunnel face, the mechanical mechanism of core rock mass reinforcement, and the development of an active control method for deep tunnel operation. Ultimately, our work is aimed at the realization of intelligent construction of deep and ultra-deep tunnels.

2. GZZ rock mass strength criterion

2.1. Scale effect of the rock mass strength criteria

Although rock strength criteria have been studied for over a century, a universal strength theory catering for all types of rocks and rock masses has yet to be proposed. As shown in Fig. 1, both the strength theory and the constitutive model have evident scaling effects. At the meso scale, the dislocation and sliding properties between mineral particles may obey simple particle contact or friction constitutive properties; however, the focus of this research is on meso-fracture mechanics. On the laboratory scale, the mechanical properties of rock are related to the macroscopic behavior of meso-rock mechanics. Although rock obeys the elastic brittleness and linear strength characteristics of MC materials, the focus of this research is primarily on the elastic deformation stage. On the engineering scale, rock mechanics reflect the properties of weak structural planes, such as joints and cracks, the mechanical strengths of which are significantly lower than that of an intact rock. This implies that the mechanical properties of an intact rock cannot be revealed on the meso or laboratory scale. The strength and deformation properties of a rock mass are chiefly represented by the statistical mechanical characteristics of disordered weak structural planes on the macro scale. These rock masses exhibit the ductile deformation and failure characteristics of elasto-plasticity and strain softening, along with evident nonlinear characteristics. At present, the focus of theoretical research is no longer limited to the elastic stage, instead extending to the plastic deformation stage of rock masses. In this context, it should be noted that the yield criterion represents the elastoplastic boundary, and is the judging standard for the rock stability and stress state, thereby rendering it essential for stability analyses and the

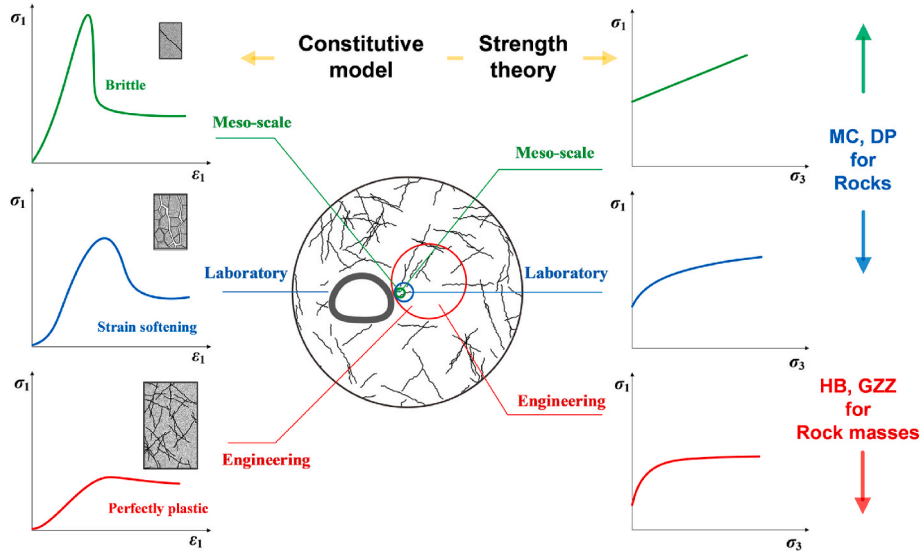


Fig. 1. Schematic representation of the size effects of the rock (rock mass) strength theory and the constitutive model.

accurate control of deep-buried surrounding rock masses.

2.2. GZZ three-dimensional and nonlinear rock mass strength criteria

The 2D HB criterion (Eq. (1)), which is widely used in engineering stability design, reflects the changes in the nonlinear characteristics of the rock mass strength with variation in the confining pressure. Its parameters can be obtained from in situ rock mass structural plane observations and indoor rock mechanics tests. Consequently, a set of complete and reliable strength parameter acquisition methods is formed.⁵⁴

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_b \frac{\sigma_3}{\sigma_c} + s \right)^a \quad (1)$$

In Eq. (1), m_b , s , and a are empirical parameters reflecting the characteristics of the rock mass, while σ_c is the intact rock uniaxial compressive strength.

Considering the complex 3D stress conditions of practical rock mass engineering, many scholars have developed different 3D HB strength criteria based on the original HB criterion.^{55–59} Among them, Zhang & Zhu proposed a generalized 3D nonlinear HB strength criterion, otherwise known as the Generalized Zhang-Zhu strength criterion (GZZ),^{57,58} by combining the advantages of the Mogi strength criterion^{60–62} and the HB criterion, yielding:

$$\frac{1}{(\sigma_c)^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right)^{1/a} + \frac{m_b}{2} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right) - m \sigma_{m,2} - s \sigma_c = 0 \quad (2)$$

where $\sigma_{m,2} = (\sigma_1 + \sigma_3)/2$, and τ_{oct} is the octahedral stress. Another version of the GZZ strength criterion that is expressed using the Lode angle and stress invariants can be expressed as:⁶³

$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\sqrt{3} J_2 \right)^{1/a} + \left(\frac{3+2 \sin \theta_\sigma}{2\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s \sigma_c = 0 \quad (3)$$

where θ_σ is the Lode angle and $\theta_\sigma = \tan^{-1} \left[\frac{2\sigma_2 - (\sigma_1 + \sigma_3)}{\sqrt{3}(\sigma_1 - \sigma_3)} \right]$. In addition, I_1 is first stress invariant, $I_1 = \sigma_{ii} = \sigma_1 + \sigma_2 + \sigma_3$; J_2 is the second invariant, and $J_2 = [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]/6$; J_3 is the third deviatoric stress invariant.

The GZZ strength criterion intersects with the HB criterion at $\theta_\sigma = -\pi/6$ and $\theta_\sigma = \pi/6$, which makes it possible for the GZZ criterion to directly adopt the HB criterion parameter system.^{57,58,64} Consequently,

the GZZ criterion has been developed as one of the recommended criteria by the International Society for Rock Mechanics and Engineering (ISRM) and the Chinese Society for Rock Mechanics and Engineering (CSRME).^{65–67}

Aimed at the non-smoothness and non-convexity of the GZZ criterion on the π plane, Zhang et al.^{63,68} modified the GZZ criterion by introducing the Lode potential function to obtain a smooth and fully convex criterion, though its complex mathematical form and plastic flow vector greatly increased the calculation overhead. Chen et al.⁶⁹ further modified the original GZZ criterion, but its mathematical form was more complicated than the original one, and its smoothness and full convexity were not mathematically proved, thereby limiting the engineering application of this criterion. Recently, Cai et al.⁷⁰ developed a new version of the GZZ criterion using a simple smoothed envelope technique based on 15 sets of true triaxial test data of rock in the literature. The geometric properties on the π -plane in terms of its smoothness and convexity were well stated, and the mathematical expression of the new criterion ($F = 0$) was defined as follows:

$$F = \frac{1}{\sigma_c^{(1/a-1)}} \left(\sqrt{3} J_2 \right)^{1/a} + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s \sigma_c = 0 \quad (4)$$

The mathematical form of Eq. (4) is very similar to the original GZZ criterion (Eq. (3)), having three intersections on the π -plane, i.e., $\theta_\sigma = \pm \pi/6$ and $\theta_\sigma = 0$. Consequently, Eq. (4) could be considered to be a further modification of the GZZ criterion. Its smooth conductivity provides convenience for finite element method (FEM) calculations, and good mechanical properties tend to be generated at higher stress levels and poor quality rock masses.⁷⁰ This equation can therefore be used for the stability analysis of deep-buried rock tunnels under high and complex stress conditions.

The non-associated flow rule is adopted in our numerical calculations. The plastic potential function (Q) could reflect the influence of the confining pressure (or dilatancy) on the volumetric strain of the rock mass,^{70,71} and the corresponding GZZ-based mathematical expression of Q can be written as:

$$Q = \frac{1}{\sigma_c^{(1/a-1)}} \left(\sqrt{3} J_2 \right)^{1/a} + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{3}} \right) m_b \sqrt{J_2} - \beta m_b \frac{I_1}{3} - s \sigma_c \quad (5)$$

where $\beta = \begin{cases} 1 - \sigma_3/\sigma_3^{cv} & (0 \leq \sigma_3 \leq \sigma_3^{cv}) \\ 0 & (\sigma_3 > \sigma_3^{cv}) \end{cases}$, which reflects the confining pressure on the rock expansion or contraction characteristics,⁷¹ and σ_3^{cv} is a confining stress beyond which the rock shows no volume change.

The plastic flow vector ($\partial Q/\partial \sigma$) of Eq. (5) can therefore be written as:

$$\begin{bmatrix} \frac{\partial Q}{\partial \sigma_1} & \frac{\partial Q}{\partial \sigma_2} & \frac{\partial Q}{\partial \sigma_3} \end{bmatrix} = \begin{bmatrix} \frac{\partial Q}{\partial I_1} & \frac{\partial Q}{\partial \sqrt{J_2}} & \frac{\partial Q}{\partial J_3} \end{bmatrix} [T]_{3 \times 3} \quad (6)$$

where,

$$\begin{cases} \frac{\partial Q}{\partial I_1} = -\frac{1}{3}\beta m_b \\ \frac{\partial Q}{\partial \sqrt{J_2}} = \frac{\sqrt{3}}{a\sigma_c^{(1/a-1)}}(\sqrt{3}J_2)^{1/a-1} + \left(\frac{\sqrt{3}}{2} - \frac{\sin 3\theta_\sigma}{\sqrt{3}}\right)m_b \\ \frac{\partial Q}{\partial J_3} = -\frac{3m_b}{4J_2} \end{cases} \quad (7)$$

and

$$[T]_{3 \times 3} = \begin{bmatrix} \frac{1}{S_1/(2\sqrt{J_2})} & \frac{1}{S_2/(2\sqrt{J_2})} & \frac{1}{S_3/(2\sqrt{J_2})} \\ S_2S_3 + J_2/3 & S_1S_3 + J_2/3 & S_1S_2 + J_2/3 \end{bmatrix}$$

The plastic flow vector $\partial Q/\partial \sigma$ can be computed by Eq. (6), and $\sin 3\theta_\sigma$ in the term of $\partial Q/\partial \sqrt{J_2}$ is zero when $\theta_\sigma = \pm\pi/6$. This confirms that the plastic flow vector of the new criterion is both continuous and derivative.

Although the 3D FEM is able to consider the hardening and softening characteristics of rock materials, the perfectly elastic-plastic constitutive model is mainly applied in this study because of the difficulties in determining the hardening and softening parameters of rock mass. Essentially, a perfectly elastic-plastic constitutive model has a good applicability for soft rock or high stress rock showing obvious ductility characteristics. Furthermore, the non-associated flow rule used in this study can reflect the effects of the confining pressure on deformation of the rock mass, and it may therefore weaken the dependence of the constitutive model in numerical calculations to a certain extent.^{71–73}

2.3. In situ intelligent acquisition of the GZZ rock mass strength parameters

The GZZ and HB criteria employ the same parameters, namely m_b , s , a or m_i , GSI , D . More specifically, m_i is an indoor rock parameter, which can be directly obtained based on the recommendations of Hoek et al.⁷⁴ In addition, the GSI is a key parameter that comprehensively reflects the engineering hydrogeology and rock mass information, and its accuracy has a direct influence on the numerical results. Furthermore, m_b is the

rock mass parameter, and D is the engineering disturbance coefficient. The two sets of parameters can be interconverted as shown below:⁷⁴

$$\begin{cases} m_b = m_i \exp\left(\frac{GSI - 100}{28 - 14D}\right) \\ s = \exp\left(\frac{GSI - 100}{9 - 3D}\right) \\ a = \frac{1}{2} + \frac{1}{6}[\exp(-GSI/15) - \exp(-20/3)] \end{cases} \quad (8)$$

The parameter system of the HB criterion (8) and the acquisition method have been widely adopted by many rock mechanics engineers and designers around the world, and its reliability and accuracy in practical engineering have been continuously verified.⁵⁴

Rock mass parameters, such as the GSI and D used in our research and calculations, originate predominantly from the tunnel site. As shown in Fig. 2, the numerical acquisition method based on a mobile phone camera or 3D laser scanning, is used to obtain the 3D point cloud information of the tunnel face, after which a 3D reconstruction method^{48,49} is employed to extract the structural geometric information of the rock mass joint fractures, such as the occurrence, space, trace, roughness, and opening degree. Finally, the geological and rock mass parameters, such as the GSI , are obtained based on the recommended methods of Hoek et al.^{52,53} This approach can quickly, automatically, and accurately give the strength parameters of the GZZ criterion, and has achieved good results in many tunnels in China. This method and corresponding research results have been reported in *Science* to be the key scientific breakthroughs of Chinese universities in 2020.⁷⁵

The HB and GZZ strength criteria are used mainly in the strength and stability analyses of rock mass on the engineering scale, while the MC strength criterion is used mainly in the stability analysis of rock materials on the laboratory scale. The equivalent parameters of the rock mass cohesion (c) and the internal friction angle (φ) used for the MC criterion can be computed based on Hoek's recommendation:⁷⁴

$$\varphi = \sin^{-1} \left[\frac{6am_b(s + m_b\sigma_{3n})^{a-1}}{2(1+a)(2+a) + 6am_b(s + m_b\sigma_{3n})^{a-1}} \right] \quad (9)$$

$$c = \frac{\sigma_c[(1+2a)s + (1-a)m_b\sigma_{3n}](s + m_b\sigma_{3n})^{a-1}}{(1+a)(2+a)\sqrt{1 + \frac{[6am_b(s + m_b\sigma_{3n})^{a-1}]}{(1+a)(2+a)}}} \quad (10)$$

where $\sigma_{3n} = \sigma_{3\max}/\sigma_c$, and $\sigma_{3\max}$ is the upper limit of the maximum

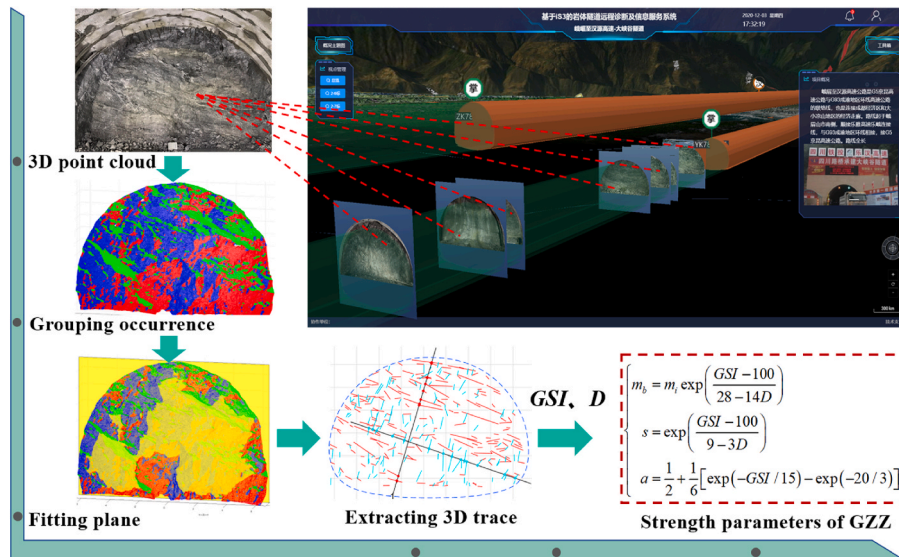
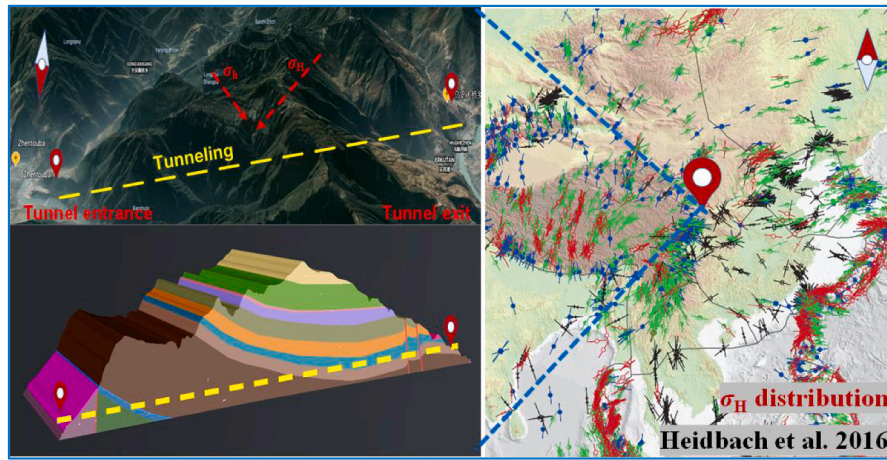
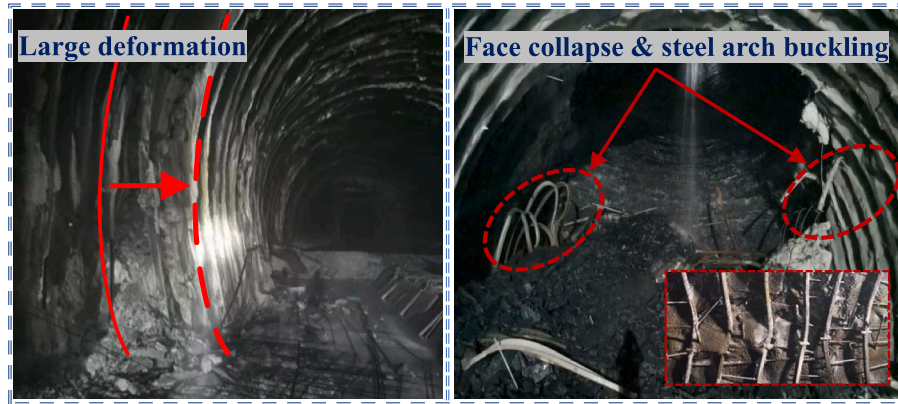


Fig. 2. Numerical processing flow for obtaining the rock mass strength parameters.



(a) Topographic map and 3D geological model



(b) Typical engineering disasters close to the tunnel face

Fig. 4. Engineering geology and typical disasters close to the tunnel face of the DXG deep tunnel.

corresponding in situ geological survey data. The angle between the tunnel axis and the maximum horizontal ground stress is 52° , and the lateral pressure coefficients in the direction of the maximum and minimum horizontal principal stresses are $\lambda_{Hv} = 1.32$ and $\lambda_{hv} = 0.874$, respectively, as determined by the in situ field measurements, the geological survey report, and the regional geo-stress distribution. The

surrounding rock is mainly unweathered or medium weathered dolomite. The intact dolomite is a hard rock with a saturated uniaxial compressive strength of $\sigma_c = 140.17$ MPa and an integrity factor of $K_v = 0.65$, as determined by indoor tests and field survey data.

Typical tunnel disasters (Fig. 4(b)) close to the tunnel face, such as high-intensity rock bursts, large deformations, and serious distortions of

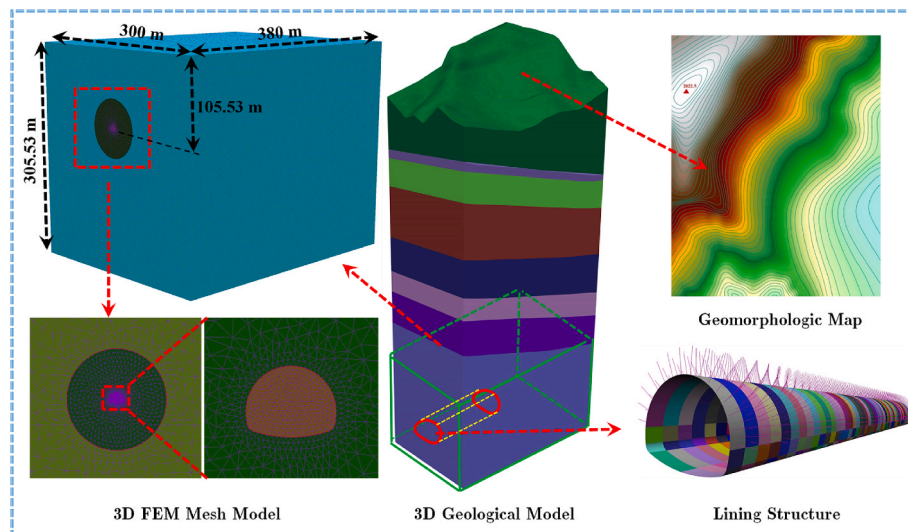


Fig. 5. 3D Numerical model of the DXG tunnel.

the steel arches, are induced by high ground stresses and occur relatively frequently. Such disasters are more severe than those occurring in shallow tunnels; however, almost no severe disasters take place in the area far from the tunnel face where the tunnel can be treated as a plane strain model. These tunnel disasters induced by the spatial effects close to the tunnel face have a significant impact on the surrounding rock stability and construction safety.

3.2. Three-dimensional numerical modeling of tunnel excavation

Fig. 5 shows the numerical model of the DXG tunnel, the tunnel face being in the x-z-coordinate axis, and the y-axis being consistent with the direction of the tunnel excavation (i.e., the tunnel axis). The geometric parameters of the tunnel were set based on the tunnel design data, the stratum, and the rock mass parameters referring to the DXG tunnel geological survey data. In the 3D FEM model, a fixed constraint was applied to the base of the model, and a load boundary was applied to the side. The calculated length of the tunnel excavation was 100 m, and the cyclic footage was 2.0 m, wherein each cyclic footage was divided into two construction steps (i.e., excavation and support). The thickness of the initial support (steel arch and anchor sprayed concrete) was 15 cm, while the anchor length was 250 cm with an equidistant distribution of 120×120 cm.

To facilitate the comparative analysis, different vertical loads were applied to simulate the tunnels of different buried depths. The dimensions of the numerical model were $300 \text{ m} \times 305.53 \text{ m} \times 380 \text{ m}$, and the model consisted of three parts (see the 3D FEM mesh model in Fig. 5 for more details), namely the tunnel, the transition circle rock outside the tunnel, and the rock formation, which permitted optimization of the calculation time to ensure the calculation accuracy. The corresponding grid sizes were 0.6, 2, and 10 m for the three parts, respectively, and the numerical model consisted of 176691 nodes and 1046938 units. This model could satisfy the requirements of refined analysis in terms of the computational convergence and results dispersion after several trial calculations.

The MC, HB, and GZZ strength criteria were used to simulate the excavation process of shallow (272 m), medium (672 m), and deep (1672 m) rock tunnels. The geometric structural information of the rock mass joints and cracks, such as the occurrence, group number, and distribution of the tunnel face, were extracted through on-site numerical acquisition technology to obtain the rock mass strength parameters.^{48,49} The rock mass parameters, such as the GSI and the D , were obtained and used for the HB- and GZZ-based numerical calculations. The equivalent strength parameters of the rock mass c and φ were computed through Eqs. (9) and (10) for use in the MC-based numerical calculations. The

strength and other mechanical parameters were obtained through in situ intelligent acquisition methods, in situ ground stress tests, rebound tests, indoor rock mechanics tests, and geological survey reports. The obtained mechanical and strength parameters for the calculation were therefore as follows: $\rho = 2650 \text{ kg/m}^3$, $E = 6270 \text{ MPa}$, $\nu = 0.28$, $\sigma_c = 140.17 \text{ MPa}$, $GSI = 46$, $D = 0.35$, $m_i = 11$, $c = 3.927 \text{ MPa}$, and $\varphi = 31.65^\circ$.

It should be noted that this equivalent method based on the 3D HB (or GZZ) strength criterion is mainly suitable for the continuous analysis of rock mass, but it is limited to tunnels with major faults or dominant joint planes. However, the rock structure parameters obtained by our suggested methods could also be used as important parameters for other analytical methods, such as discontinuous analysis.^{49,78} In addition, the lithology of the rock in DXG tunnel is relatively homogeneous as a whole and the occurrence is approximately horizontal. Thus, the GZZ-based continuous method was able to generate reliable results.

3.3. Longitudinal displacement profiles (LDPs) of the surrounding rock

Fig. 6(a)–(c) shows cloud diagrams for the surrounding rock convergent deformation for the deep rock tunnel (GZZ-based). The crown settlement, sidewall horizontal displacement, and tunnel face extrusion deformation were all between 95 and 100 mm, while the floor heave deformation was ~ 150 mm. Fig. 6(d)–(e) show the stress and displacement cloud diagrams of the initial support structure, wherein the σ_1 value generally ranges from 150 to 250 MPa, but sometimes reaches >500 MPa due to stress concentration; the maximum displacement exits at the bottom of the arch structure and has a value of 158.6 MPa, which is consistent with the deformation of the tunnel floor (Fig. 6 (b)). However, the stress and displacement of the surrounding rock close to the tunnel face (Fig. 6(d)–(e)) are relatively small compared with those far away from the tunnel face area, and sharply trend to the maximum (or ultimate) value, thereby demonstrating a clear 3D spatial effect close to the tunnel face.

Fig. 7 shows LDPs of the surrounding rock convergent deformation (i.e., the crown settlement, floor heave, and sidewall horizontal displacement) during the excavation using different strength criteria. The MC-based numerical results exhibited slight differences to the HB-based results due to the fact that the rock mass c and φ values used for the MC criterion were equivalently derived from the HB parameters, which proved to be reliable for this approach.

In the original design plan, the inverted arch controlling the deformation of the floor heave was used as the secondary lining for safety reserves, and its supporting strength and rigidity were low, resulting in greater deformation of the floor heave (red curve, Fig. 7). The steel arches used to prevent crown and sidewall deformation had a higher

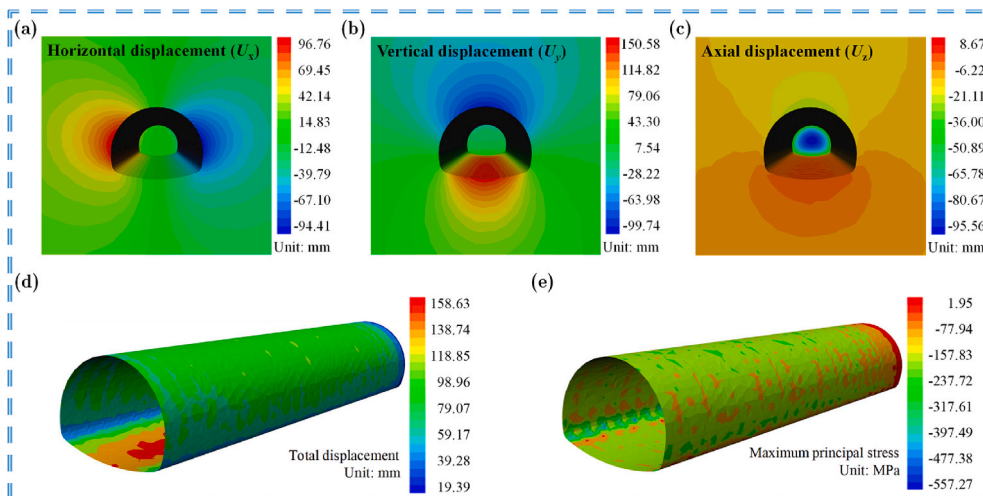


Fig. 6. Cloud diagrams of the surrounding rock convergent deformation.

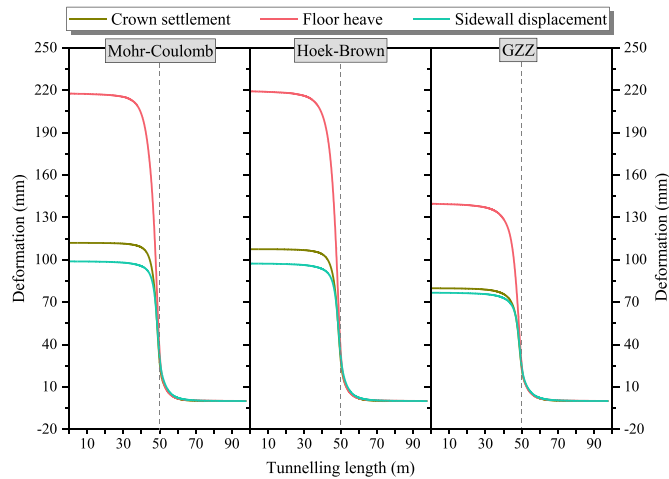


Fig. 7. LDPs of the surrounding rock based on different strength criteria.

radial rigidity, and so exhibited good results in terms of restricting the deformation. Consequently, in the design and construction of deep tunnels, inverted arches should be closed timeously and have sufficient strength and stiffness; it is unreasonable to use them as a safety reserve support or secondary lining.

The deformation of the floor heave calculated based on the 2D MC

and HB criteria was 1.6 times that of the GZZ criterion, wherein the crown and sidewall deformations were 1.4 times that of the GZZ criterion, thereby reflecting the influence of σ_2 on the strength and deformation. These results demonstrated that use of 2D criteria without considering σ_2 is not appropriate for deep rock tunnels that exhibit complex 3D stress conditions. Moreover, the consistent results of the two 2D criteria (i.e., the linear MC and nonlinear HB criteria) and the difference between the HB- and GZZ-based results indicate that the 3D issue related to the rock mass strength criteria is more significant than the nonlinear issue. It should also be noted that the rheological deformation is not considered and the computed deformation is mainly caused by the spatial effect of tunnel excavation because of the high-quality rock and short construction period that show the insignificant rheological behavior.⁷⁹

3.4. Reliability verification of the GZZ criterion using field deformation monitoring data

The dolomite of the DXG deep tunnel is of good quality, and so severe rock bursts and tunnel face failures can be created during excavation under high geostress conditions. According to the numerical results, the displacement release coefficient (DRC) of the tunnel crown is generally 25–35% at the tunnel face section, 50–70% at section 2 m away from the tunnel face, and >80% at the section further than 4 m away from the tunnel face. Considering the safety and construction risks, the displacement monitoring points were situated at a section 3–7 m away

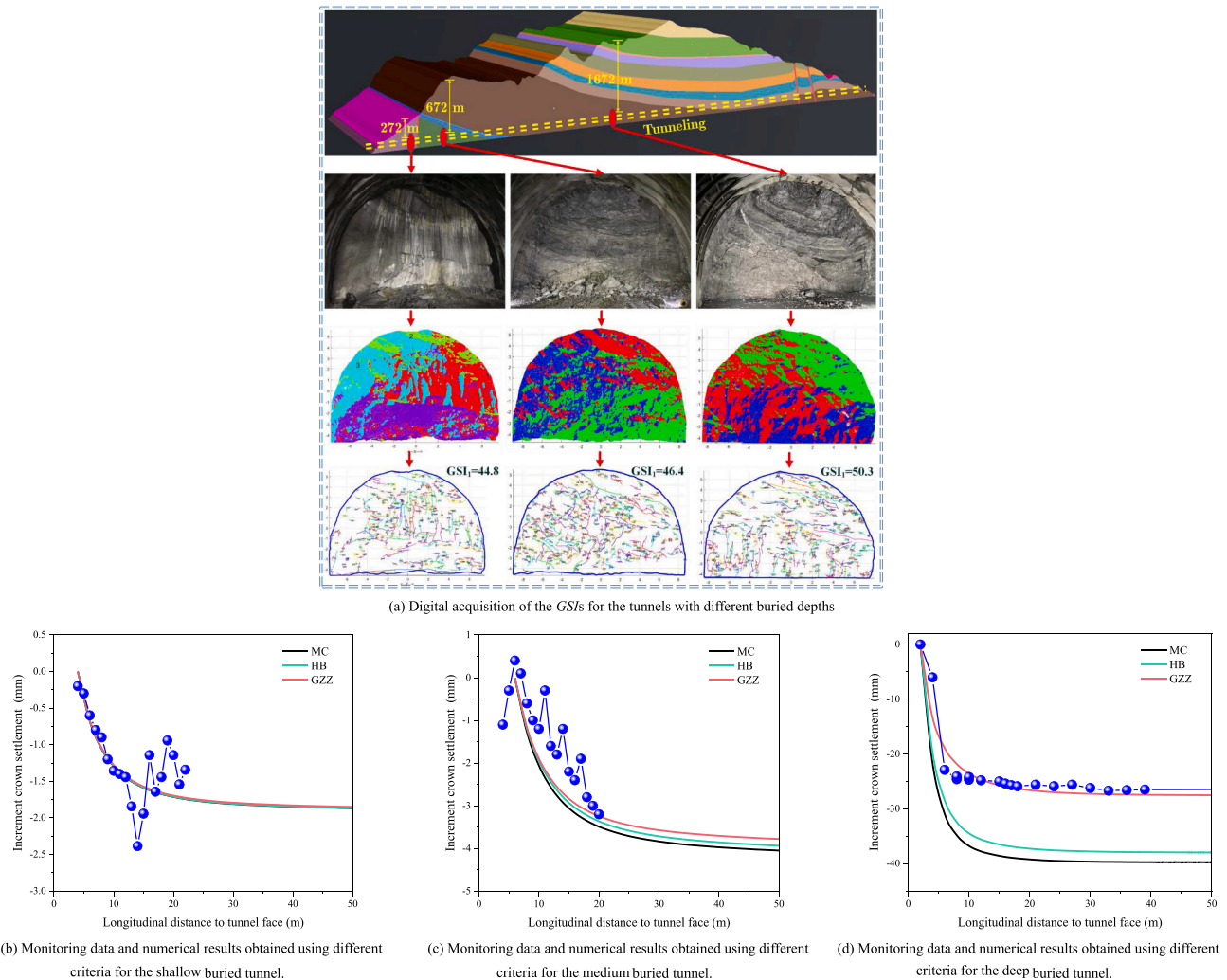


Fig. 8. Monitoring data and numerical results for the different buried tunnels.

from the tunnel face, so the monitoring values may not reflect the actual total deformation of the tunnel due to the high displacement release coefficient close to the tunnel face. Consequently, a deformation increment was used to compare the numerical results of different strength criteria.

Subsequently, method verification was carried out for a shallow buried tunnel (252 m), a medium buried tunnel (672 m), and a deep buried tunnel (1672 m), as presented in Fig. 8. The data used for verification were the increment deformation data based on the calculated values (i.e., the total deformation) minus the released part before monitoring. The data for the shallow, medium, and deep buried tunnels were initially measured at distances of 4, 4, and 2 m away from the tunnel face, respectively. Fig. 8(a) show the geological model of the DXG tunnel and the digital process of the GSI_s at different buried depths. The GSI_s for typical shallow, medium, and deep tunnels were determined to be 44.8, 46.4, and 50.3, respectively. It can be seen that the GSI gradually and slightly increases with the increase of buried depth, and this trend was related to the fact that the structural plane of the rock mass in shallow tunnels is more developed than that in deep tunnels.

Fig. 8(b) shows slight differences in the crown settlement results of the shallow tunnel when these results were obtained using different strength criteria, although the field monitoring data are in good agreement with the numerical results. In a shallow tunnel, the surrounding rock is mainly governed by the elastic stress because of the good quality rock mass, and the elastic deformation of the surrounding rock is independent of the strength criterion. Thus, the FEM calculation time for a shallow tunnel is significantly shorter than that required for a deep tunnel where a plastic zone is widely distributed. For the medium buried tunnel (Fig. 8(c)), the crown settlements calculated using the different criteria were slightly different, with the GZZ-based displacement being significantly closer to the monitoring data, in addition to being slightly smaller than the values obtained using the HB and MC criteria. The small differences between the 3D and 2D strength criteria suggested the small range of the plastic zone produced during medium tunnel excavation.

However, the deformations of the deep buried tunnel (Fig. 8(d)) based on the different criteria were observed to be very different because of the widely distributed plastic zone produced during deep tunnel excavation. The deformations of the 2D strength criteria (i.e., HB and MC) were almost twice that of the 3D criterion (i.e., GZZ), and the GZZ-based results generally agreed well with the monitoring data. Since the GZZ criterion could be completely transformed to the HB criterion under conventional stress states (i.e., triaxial tension and triaxial compression), the large difference between the GZZ and HB results also reflects the significant influence of σ_2 on the strength and deformation of the rock mass.

Since a shallow tunnel is mainly governed by the elastic stress, the differences among the different strength criteria are therefore insignificant, and so the selection of a reasonable strength criterion may often be ignored. However, the plastic zone is widely distributed in a deep tunnel, and the surrounding rock strength and deformation show a close dependence on the strength criterion. A reasonable selection of 3D and nonlinear strength criteria (e.g., GZZ), is therefore essential for deep tunnel design and control.

4. Three-dimensional and nonlinear extrusion effects of the tunnel face

It was observed that engineering disasters close to the excavation face of the deep DXG tunnels occurred more frequently and were more severe than those of the shallow tunnel due to the significant 3D spatial effects and extrusion deformation. Consequently, the simple plane strain model cannot be used in the design and construction of deep tunnels, thereby indicating the necessity to fully understand the 3D space effect of the tunnel face.

It should be noted that there will be a certain error for the computed extrusion deformation within a certain range (e.g., $y < 20$ m) close to the

model boundary due to the influence of the model boundary effect. We therefore decided to mainly focus on the mechanical behaviors at the middle position of the tunnel model (i.e., $y = 50$ m) to render the analysis results more representative.

4.1. Elastoplastic stress conditions of the tunnel face under excavation unloading

The red curve in Fig. 9 depicts the extrusion deformation and stress state of the tunnel face center of the tunnels at different buried depths. The face extrusion deformations of the shallow, medium, and deep tunnels were 4.74, 16.8, and 93 mm, respectively, and the deformation curves at the buried depths exhibited diverging nonlinear trends. The histogram in Fig. 9 shows that the ratio of plastic deformation to the total deformation gradually increases with increasing buried depth. Thus, the shallow tunnel was chiefly in an elastic state with a plastic deformation proportion of 1.41%; the plastic deformation of the medium tunnel face increased sharply with the proportion of plasticity being 58.1%; and the plastic ratio of the deep tunnel was 77.9%, which accounted for the significant nonlinear characteristics of the extrusion deformation at the buried depths and the large deformation of the deep tunnel face.

It was found that the stress level (I_1) gradually increased as the buried depth increased, and the GZZ envelope on the π plane gradually expanded. In addition, the stress state of the rock element in the center of the tunnel face gradually transferred from inside the GZZ envelope (elastic condition) to outside the envelope (plasticity or failure condition). The FAI_{GZZ} for the center of the tunnel face was < 1.0 for the shallow tunnel due to the high strength of the rock (140.17 MPa), which was in an elastic stress state; however, σ_3 was equal to -0.028 MPa, which is less than zero, thereby indicating tensile stress. Furthermore, the FAI_{GZZ} for the medium tunnel was 1.119, which represents an elastoplastic stress state, and σ_3 was equal to -0.03 MPa. Moreover, the FAI_{GZZ} for the deep tunnel was 2.07, which is > 2.0 , thereby indicating a plastic stress or failure condition, while σ_3 was equal to -0.06 MPa. The σ_3 of the core rock reflects a tensile value in deep tunnel, which became more pronounced in the deep tunnel, and so the unloading effect (i.e., strong disturbance) of the deep tunnel was more obvious, and the tunnel face was in an extremely unstable stress state (i.e., tensile strength ≈ 0).

4.2. Extrusion deformation of the tunnel face

Fig. 10(a) shows the U_y displacement distribution of the deep tunnel and the corresponding LDP curve in front of the tunnel face. As can be seen, the pre-extrusion deformation changed rapidly close to the tunnel

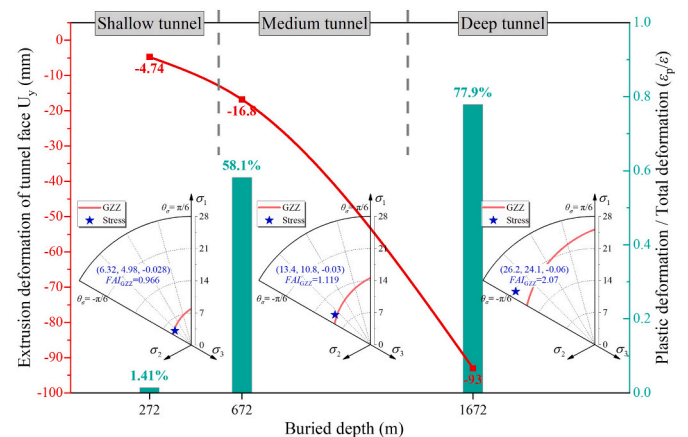


Fig. 9. 3D extrusion deformation and stress conditions of the tunnel faces at different buried depths.

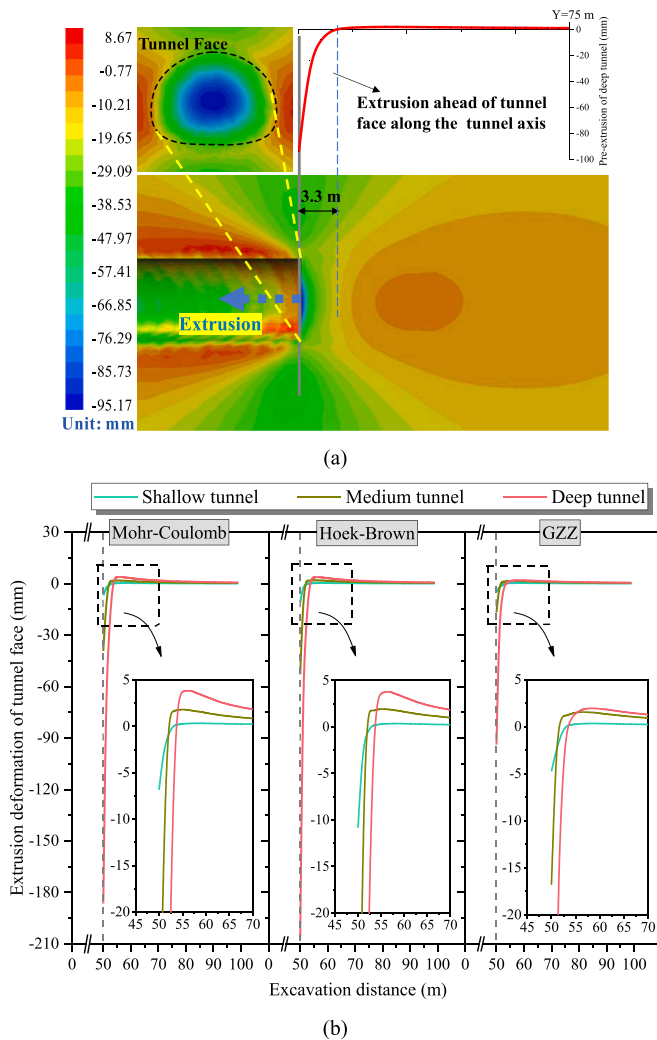


Fig. 10. Extrusion deformation of the deep tunnel face (U_y).

face, while the extrusion deformation decreased to ~ 0 within 3.3 m of the front of the tunnel face, showing a certain degree of reverse deformation. Consequently, the range of 3–4 m in front of the tunnel face can be treated as the excavation unloading influence range (or the excavation disturbance area), which is the rock mass within the disturbance area that has a significant 3D space effect.

Fig. 10(b) shows the LDP curve of extrusion deformation in front of the tunnel face, which was calculated at varied buried depths and different strength criteria, and using $Y = 50$ as the position of the tunnel face. The variations observed between the different criteria-based results in the shallow tunnel were small because of the elastic stress state of the shallow tunnel face. However, the extrusion deformation of the deep tunnel face increased significantly. More specifically, the results based on the 2D criteria (i.e., HB and MC) were 2–3 times that of the 3D criterion (GZZ), showing that σ_2 exhibited obvious restraining effects against tunnel extrusion deformation.

Since the deep tunnel exhibited significant 3D nonlinear face extrusion effects, the design and construction of a deep tunnel should pay the appropriate attention to these effects, thereby further demonstrating that deep tunnel design cannot make use of the plane strain model and design ideas.

4.3. Distribution of unstable zone in the deep tunnel and its evolution during the excavation

Extrusion deformation is the most important feature of the 3D spatial effect of a deep tunnel. Fig. 11 depicts the regional distribution of $YAI_{GZZ} > 2.0$ for a deep tunnel at different construction steps, wherein the observation section correlates with $Y = 50$. The dotted line in Fig. 11 represents the tunnel outline; the edge of the red area is the contour line of $YAI_{GZZ} = 2.0$; the area within the red line is $YAI_{GZZ} > 2.0$ and the area outside the area is $YAI_{GZZ} < 2.0$. The transfer law of $YAI_{GZZ} = 2.0$ under different construction steps can be used to evaluate the stability mechanism of the surrounding rock during excavation.

During the excavation, the plastically unstable zone initially appeared in a certain range in front of the tunnel face (e.g., construction step $T = 48$ in Fig. 11), wherein the unstable zone was distributed in the center of the face. At the tunnel face ($T = 50$), the $YAI_{GZZ} = 2.0$ area gradually expanded and the regional shape was similar to the tunnel outline. It is worth noting that the tunnel face rock close to the tunnel wall remained at $YAI_{GZZ} < 2.0$, and so was more stable than the center of the tunnel face. As the excavation continued, a red area ($YAI_{GZZ} > 2.0$) appeared at the base of the tunnel behind the tunnel face, but did not appear in the sidewall area. This explains why the floor heave deformation was greater than the crown settlement and sidewall horizontal deformations shown in Fig. 7. Furthermore, the stress and deformation of the $Y = 50$ section can be considered to be in their final state at $T = 60$, where there was a large area of $YAI_{GZZ} > 2.0$ distributed on the tunnel face and tunnel floor, and a smaller area of $YAI_{GZZ} > 2.0$ distributed at the tunnel crown and sidewall because of the high radial rigidity of the steel arch.

It can be seen that a plastic (failure) zone initially appeared in front of the tunnel face, which may also explain why the engineering disasters of the DXG deep tunnel occurred more frequently and severely close to the tunnel face (Fig. 4(b)). With the advancement of excavation, the surrounding rock behind the tunnel face became increasingly unstable. It can therefore be considered that the tunnel face instability originated at an earlier point than that of the surrounding rock, and that the tunnel face was the most important tunnel stability factor in the deep tunnel. This conclusion is consistent with the viewpoint of Lunardi et al.^{13,18}

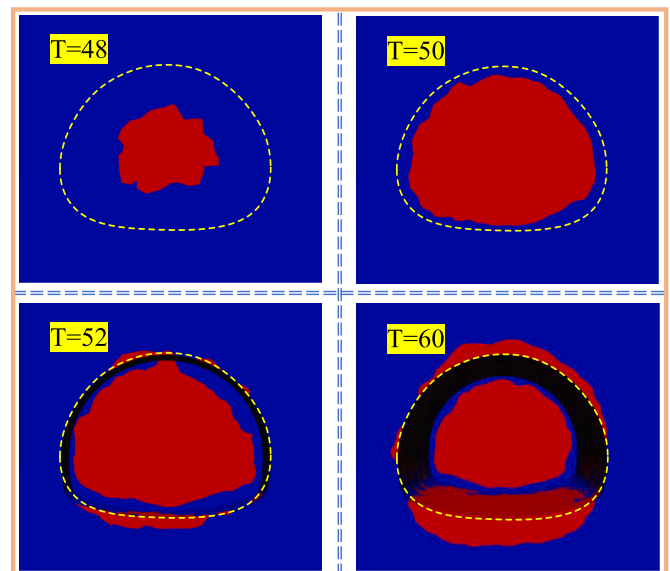


Fig. 11. Regional distribution of $YAI_{GZZ} > 2.0$ for a deep tunnel at different construction steps (observation section: $Y = 50$).

5. Reinforcement effects of the core rock mass in the deep tunnel

5.1. Determination of the reinforcement range and parameters

Since the rock mass in front of the tunnel face initially appears unstable, the extrusion deformation of the tunnel face, which is known to be the most notable feature of 3D nonlinear spatial effects, is an important factor in terms of influencing of the stability of deep tunnels. This study therefore discusses the mechanism by which the core rock mass reinforcement influences the stability of the surrounding rock. There are many ways to strengthen the core rock mass in front of the tunnel face, such as advanced pipe roofing, jet grouting, and glass fiber anchoring. Based on the 3D FEM calculations, the reinforcement effect is based on an improvement of the mechanical strength, stiffness, and stability of the advanced core rock no matter what type of reinforcement method is selected. The aforementioned analysis determined the strong disturbance area in front of the tunnel face to be 3–4 m, wherein the reinforcement range was 4 m. The GSI is the core strength parameter of the GZZ criterion and is directly related to the rock strength and mechanical parameters.⁸⁰ Therefore, the GSI was selected as the reinforcement parameter. The GSI values of the core rock mass were therefore set as $GSI_0 = 46$, $GSI_1 = 51$, $GSI_2 = 56$, $GSI_3 = 61$, and $GSI_4 = 66$.

(1) GZZ strength parameters (m_b , s , a)

The strength parameters of the HB and GZZ criteria are calculated using Eq. (8).

(2) MC strength parameters (c , φ)

The strength parameters of the MC criterion (rock mass c and φ) are calculated using the equivalent method with Eqs. (9) and (10).

(3) Rock mass elastic modulus

The elastic modulus (E_m) and Poisson ratio (ν_m) are often used as the main mechanical parameters for rock mass, where ν_m is a dimensionless decimal that does not consider changes in the GSI . As outlined in Eq. (16), Hoek et al. obtained E_m based on engineering data and experience:⁸¹

$$E_m = 100000 \left[\frac{1 - D/2}{1 + e^{(75+25D-GSI)/11}} \right] \quad (16)$$

Here, we cannot directly compute the empirical value of E_m based on Eq. (16) because it may not completely match the actual value obtained from the geological survey data and the in situ field tests, but it is assumed that E_m follows the change law governed by Eq. (16). Thus, Eq. (16) can be rewritten as:

$$E_m = k \left[\frac{1 - D/2}{1 + e^{(75+25D-GSI)/11}} \right] \quad (17)$$

where k is a constant and Eq. (16) satisfies the initial condition, i.e., $E_{m0} = 6270$ when $GSI_0 = 46$.

The ratio of the E_m before and after reinforcement (E_{mk}/E_{m0}) can be written as:

$$\frac{E_{mk}}{E_{m0}} = \frac{1 + e^{(75+25D-GSI_0)/11}}{1 + e^{(75+25D-GSI_k)/11}} \quad (18)$$

where E_{mk} is the elastic modulus of the rock mass after reinforcement, $k = 0, 1, 2, 3, 4$; E_{m0} is the initial elastic modulus and $E_{m0} = 6270$ MPa (obtained from geological survey data).

E_{mk} can therefore be obtained from Eq. (18):

$$E_{mk} = \left[\frac{1 + e^{(75+25D-GSI_0)/11}}{1 + e^{(75+25D-GSI_k)/11}} \right] E_{m0} \quad (19)$$

The values of E_{mk} under different GSI_k can then be obtained using Eq. (19).

(4) Uniaxial compressive strength of the rock (σ_{ci})

The rock mass σ_c obtained from the HB and MC strength criteria can be expressed as follows:

$$\sigma_{cm} = \begin{cases} \frac{2c \cos \varphi}{1 - \sin \varphi} & \text{MC} \\ s^a \sigma_{ci} & \text{HB} \end{cases} \quad (20)$$

Although the rock σ_{ci} value is one of the strength parameters of the GZZ (HB) criterion, the result from Eq. (20) is that of the rock mass σ_{cm} . Considering that the MC criterion is used mainly for indoor rock materials, the MC method can be used to predict the rock σ_{ci} under different GSI_k values. Using a similar method to obtain the E_m of the rock mass, the σ_{ci} under different GSI_k values can be obtained as follows:

$$\sigma_{ck} = \frac{2c_k \cos \varphi_k / (1 - \sin \varphi_k)}{2c_0 \cos \varphi_0 / (1 - \sin \varphi_0)} \sigma_{c0} = \frac{c_k \cos \varphi_k \cdot (1 - \sin \varphi_0)}{c_0 \cos \varphi_0 \cdot (1 - \sin \varphi_k)} \sigma_{c0} \quad (21)$$

Based on the above analysis, the main rock mass strength and mechanical parameters can be obtained, and are listed in Table 1.

5.2. Extrusion deformation and stress state of the tunnel face under different GSI values

Table 2 and Fig. 12 describe the extrusion deformations and stress conditions of the deep tunnel face under different reinforcement parameters (i.e., the different GSI s). The stress level and GZZ envelope on the π plane gradually expand with an increasing GSI , while the stress state (blue points in Table 2 on the π plane) gradually changes from outside to inside the envelope as the GSI increases. YAI_{GZZ} gradually decreases from $YAI_{GZZ} = 2.07$ (the yield or failure condition of the unreinforced core rock mass with $GSI = 46$) to $YAI_{GZZ} = 0.97$ (the elastic stress condition of the reinforcement rock mass with $GSI = 66$), thereby indicating that the stability and mechanical properties of the tunnel face gradually improved as the GSI increased. Moreover, the rock element in the tunnel face center changes from tensile stress to compressive stress, σ_3 gradually increases, and the unloading effect is reduced.

Reinforcement reduces the stress unloading effect of the core rock mass and increases the stress level remaining inside the rock mass. It is also known that the advance core rock mass plays the most significant role in supporting the pre-convergent deformation of the surrounding rock mass, wherein the reinforcement increases the σ_3 value of the surrounding rock, and so greatly enhances the strength of the core rock mass based on the changing characteristics of the rock mass strength with variation in σ_3 . Consequently, the reinforcement gives full play to the strength potential of the rock material, rendering the rock itself a natural supporting material with a superior natural support in terms of controlling the pre-convergent deformation of the surrounding rock.

Fig. 12 shows an obvious non-linear change in the tunnel face extrusion deformation with a variation in the GSI (red curve). More specifically, it can be seen that the reinforcement has an important influence in restraining the extrusion deformation when the GSI is low (i.e., soft rock), but the reinforcement effect is not sensitive when the GSI is high (i.e., hard rock). When the $GSI = 66$, the rock element of the tunnel face center is in an elastic stress state with a YAI_{GZZ} of 0.97, and it can be inferred that the reinforcement effect will not be effective when the GSI is > 66 , and this may result in wasted reinforcement materials. When the GSI is < 61 , YAI_{GZZ} presents a convex shape with changes in the GSI (the dotted line in Fig. 12), while a concave shape is observed when the GSI is > 61 . Therefore, a GSI of 61–66 can be taken as the optimal

Table 1
Strength and mechanical parameters of deep tunnels under different GSI_k values.

GSI_k	Strength parameters							Mechanical parameters	
	GSI	σ_{ci} (MPa)	m_b	s	a	C (kPa)	Φ (°)	E_m (MPa)	ν_m
GSI_0	46	140.17	1.06	0.001122	0.5076	3927	31.65	6270	0.28
GSI_1	51	160.80	1.32	0.002105	0.5054	4341	33.44	9703	0.28
GSI_2	56	184.88	1.64	0.003948	0.5038	4808	35.21	14872	0.28
GSI_3	61	213.86	2.03	0.007405	0.5026	5357	36.95	22470	0.28
GSI_4	66	249.96	2.52	0.013888	0.5018	6031	38.65	33253	0.28

reinforcement parameter for the advance core rock of the DXG deep tunnel.

5.3. Influence of reinforcement on the surrounding rock deformation

By comparing the YAI_{GZZ} values under different GSI s, the reinforcement influence of the advance core rock mass on the surrounding rock can be quantitatively evaluated. As shown in Fig. 13(a), the YAI_{GZZ} close to the tunnel face changes significantly, further verifying the existence of the 3D spatial effect at the tunnel face. The impact of reinforcement on the rock tunnel face is most obvious because it is directly reinforced.

At the tunnel face, the YAI_{GZZ} exhibits the poorest stability compared with the surrounding rock, while the rocks around the tunnel (i.e., at the crown and the sidewall) are still in an elastic stress state ($YAI_{GZZ} < 1.0$). For the surrounding rock that is far behind the excavated face, YAI_{GZZ} is larger, but the large radial stiffness and strength of the steel arches and initial support play a role in preventing the radial deformation of the surrounding rock. However, the rock mass of the tunnel face is often in an unsupported state, which may result in instability or collapse of the tunnel face, especially for a deep tunnel that exhibits large extrusion deformation under high geostresses, i.e., its reinforcement has a significant influence on the stability of the tunnel face as well as the surrounding rock. It should be noted that reinforcement seems to have a limited effect on the ultimate radial deformation of the surrounding rock, but helps to stabilize the rock during the construction process, particularly close to the tunnel face.

The displacement release coefficient (DRC) is defined as the percentage of the current convergence deformation to the final convergence deformation (i.e., the maximum convergence deformation). The DRC at the tunnel face section is commonly used to evaluate the stability of the surrounding rock in the construction process close to the tunnel face. In general, a smaller DRC corresponds to a more stable tunnel face and surrounding rock during construction, indicating a reduced likelihood of disaster close to the tunnel face. Fig. 13(b) shows the DRC at the tunnel face under different GSI s. The DRC of the crown settlement and sidewall horizontal displacement were 20–30% before reinforcement; these values are consistent with engineering experience and the theoretical results of previous studies. The DRC of the floor heave was ~15%, meaning that ~85% of the deformation would be produced during the subsequent excavation, wherein the final deformation and unstable tunnel floor area are greater than the crown settlement and the sidewall. Although the smaller DRC of the floor heave was beneficial to the stability of the surrounding rock, it would be necessary to close the tunnel floor and ensure the support strength of the inverted arch as early as possible to control the large deformation of the floor heave and the risk of surrounding rock instability caused by the spatial effects of the subsequent excavation.

Fig. 13(b) also shows that the reinforcement has a great influence on the DRC of the tunnel face. The DRCs of the crown settlement and sidewall horizontal displacement were ~15% when the GSI was equal to 66, and the DRC of the floor heave was ~6%. Consequently, the convergence and pre-convergence deformation in front of the face were greatly reduced, and the stability of the tunnel face and surrounding rock was increased. The DRC curves under different GSI s show certain nonlinear trends. The reinforcement effect is obvious for poor quality

rock mass (i.e., for a small GSI), and the curves become flat for good quality rock mass (i.e., for a large GSI). It can therefore be inferred that the effect of the reinforcement on the DRC would be limited when the $GSI > 66$, verifying the aforementioned inference that a GSI of 61–66 could be taken as the optimal reinforcement parameter.

5.4. Influence of reinforcement on the convergence-constraint curve

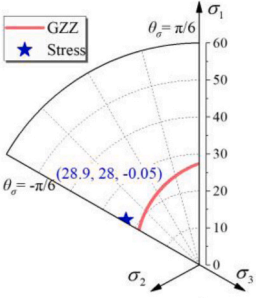
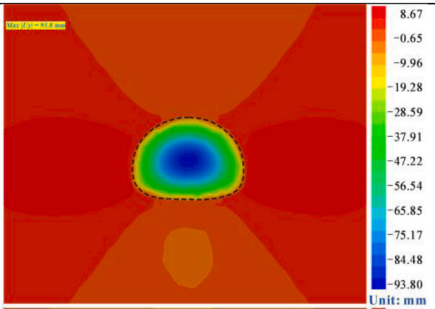
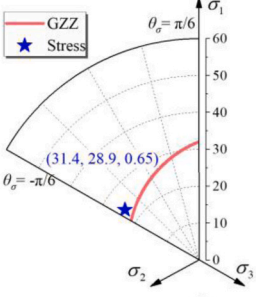
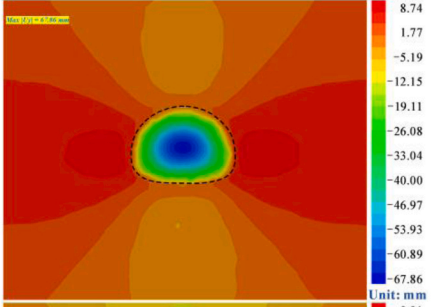
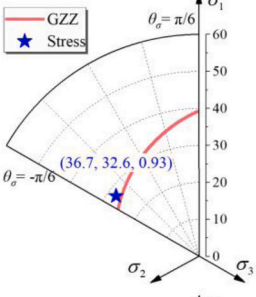
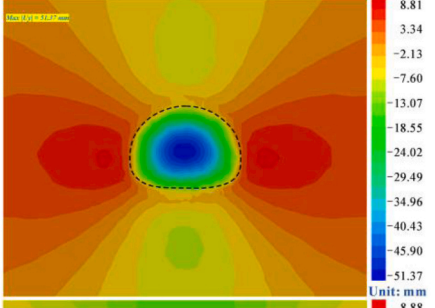
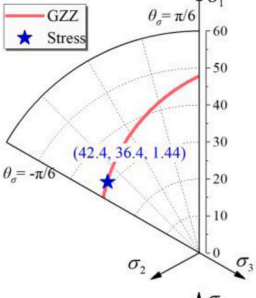
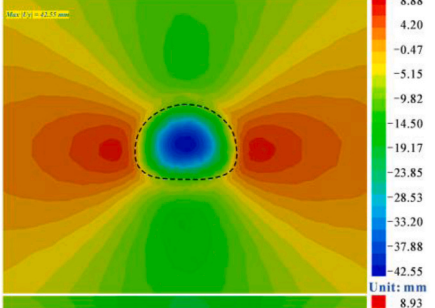
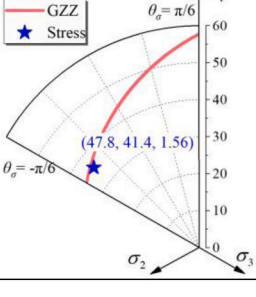
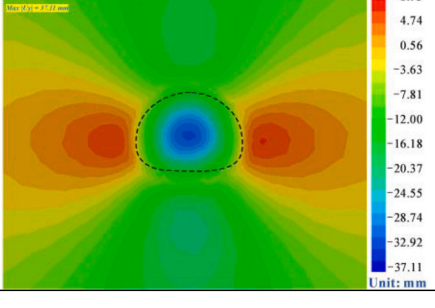
Fig. 14(a)–14(d) show the convergence-constraint (P_r-U_r) curves of different sections in front of the tunnel face. The P_r-U_r curve of the core rock mass is shown in Fig. 14(a), wherein it can be seen that the unreinforced rock curve (black curve) has obvious non-linear characteristics, and gradually transforms itself into a linear curve as the GSI increases, thereby indicating that deformation had been effectively controlled. The red dotted line represents the P_r-U_r at the tunnel face under different GSI s, which also confirms that extrusion of the tunnel face was greatly reduced after reinforcement. Fig. 14(b)–14(d) show the P_r-U_r curves of the convergence deformation along the radial direction of the tunnel in front of the tunnel face, wherein the curves of the tunnel crown, floor, and sidewall are similar. As can be seen, the unreinforced rock convergence-constraint curves are particularly steep since the severe unloading effect results in a slight radial pressure (P_r) and a large radial deformation (U_r); however, for the reinforced core rock mass, the displacement and unloading effect is limited, and the majority of the stress remains in the rock mass, exhibiting a good supporting effect on the surrounding rock mass. Consequently, the pre-convergent deformation is smaller, and the P_r-U_r curve becomes gentler after reinforcement as the GSI increases.

It should be noted that the red dashed lines in Fig. 14(a)–(d) are essentially straight curves, which indicates that the reinforcement effect results in the rock mass in front of the tunnel face exhibiting larger radial stresses and smaller radial displacements; it can also be inferred that the connection lines of P_r-U_r under different GSI s in other sections are also essentially linear curves, thereby indicating their good uniformity. This linear change law can be used to quantitatively evaluate the reinforcement effect of the advanced core rock in front of the tunnel face.

Fig. 14(e)–(g) describe the relationship curves between the pre-extrusion deformation (U_y) of the core rock and the pre-convergent deformation (U_r) of the tunnel circumference in front of the tunnel face under different GSI s. The pre-extrusion and pre-convergence deformations show positive correlations, wherein the larger pre-extrusion deformation corresponds to a larger pre-convergence deformation. The curves for the tunnel crown, floor, and sidewall all show certain non-linear characteristics. More specifically, the pre-convergent and pre-extrusion deformations are significantly reduced as the GSI increases, and it should be noted that the overall trends of the curves under different GSI s are very similar, with only small differences being observed between the different curves. Consequently, the reinforcement has a positive effect on controlling the extrusion deformation of the tunnel face and the pre-convergence deformation of the surrounding rock, but it seems to have little effect on the curve trends, wherein the relationship between the pre-extrusion and pre-convergence deformations exhibits a good uniformity even under different GSI s.

It should also be noted here that the corresponding relationship between the reinforcement measures, the reinforcement parameters, and

Table 2
Extrusion deformation and stress conditions of the tunnel face before and after reinforcement.

GSI	Stress conditions	FAI_{GZZ}	Extrusion deformation of the tunnel face
46		2.07 (Plastic)	
51		1.94 (Plastic)	
56		1.67 (Plastic)	
61		1.17 (Plastic)	
66		0.97 (Elastic)	

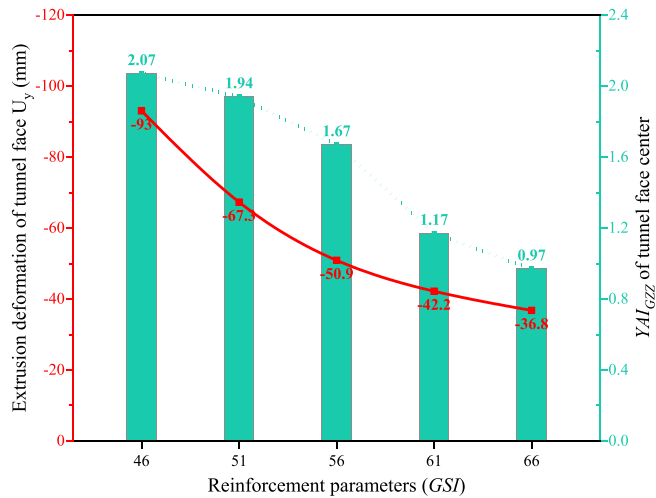


Fig. 12. Extrusion deformation and YAI_{GZZ} of the tunnel face center under different GSIs.

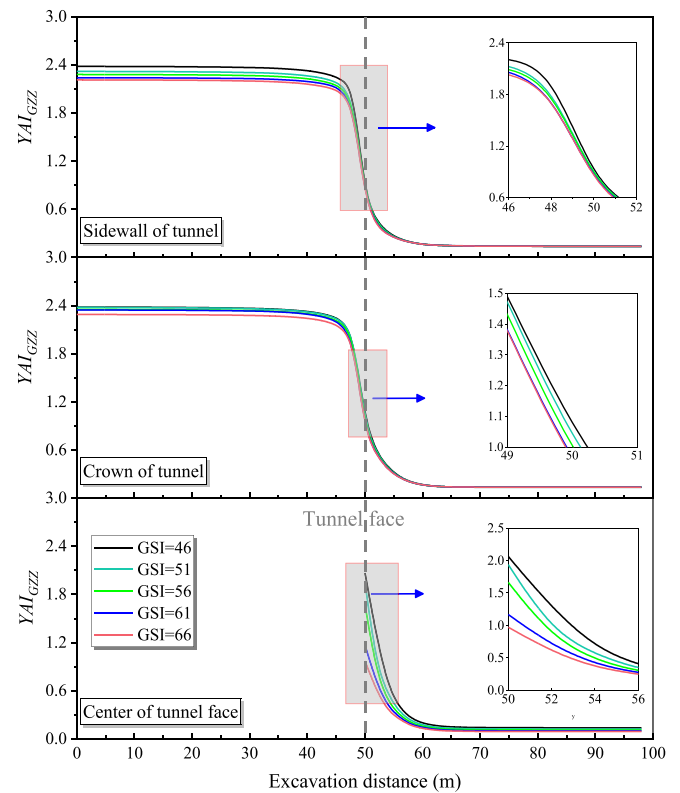
the GSI values will be further analyzed through material experiments and field experiments in our follow-up research.

6. Discussion on the active control of the rock stability in a deep rock tunnel

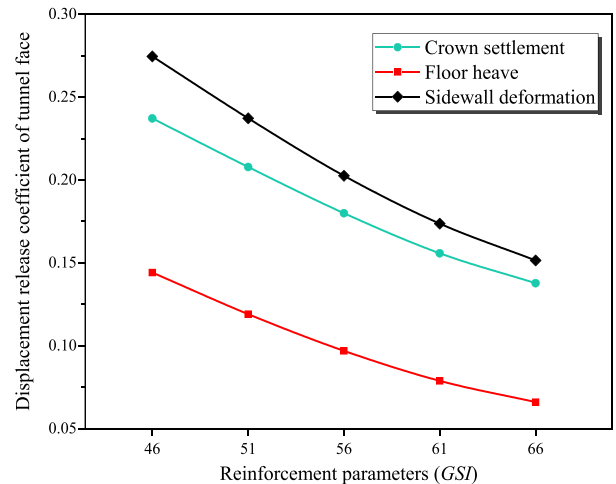
Deep rock tunnel engineering is confronted with complex 3D stress states because of the unloading and redistribution of stresses caused by strong disturbances of the deep excavation rock mass, and the obvious 3D spatial effects close to the deep tunnel face require the use of 3D nonlinear strength criteria for their refined simulation and analysis. The GZZ strength criterion and strength parameter (i.e., the GSI and D values) intelligent acquisition methods were used in our study to realize the 3D forward analysis of deep tunnel excavation with the aim of providing scientific and technical references for the intelligent construction of deep and ultra-deep tunnels.

Tunnel design and construction under plane strain assumptions usually emphasize the radial deformation of surrounding rock, ignoring the 3D pre-extrusion and pre-convergence deformation of the core rock mass. Compared with deep tunnels, shallow tunnels that accentuate plane problems exhibit a smaller extrusion deformation of the tunnel face because of their low stresses, and the designed supporting systems (such as steel arches) often possess an excellent radial stiffness and strength (ignoring the importance of the longitudinal stiffness and the overall stiffness). However, a supporting system with a low longitudinal stiffness is obviously insufficient in the face of the high stresses and large deformation along the tunnel axis in a deep tunnel, which presents obvious 3D space effects. Consequently, large radial and axial support stiffnesses should be simultaneously considered in a deep tunnel, such as increasing the width of the steel arch and strengthening its longitudinal connection.

The reinforcement of the advanced core rock mass can be regarded as an active control method for stabilizing the surrounding rock and tunnel face. As shown in Fig. 15, a strong disturbance of a deep tunnel caused by excavation unloading results in the stresses remaining in the rock mass being relatively low for the unreinforced core rock mass. The small value of σ_3 corresponds to the low strength of the rock mass (point A in Fig. 15), which produces large plastic deformation under high stresses, and large extrusion deformation of the tunnel face. In addition, the σ_1 or stress level (I_1) of the core rock mass represents the supporting stress (σ_3) of the unexcavated surrounding rock mass, i.e., the unreinforced core rock mass has a weak supporting effect against the pre-convergent deformation, resulting in a large DRC value at the tunnel face. Conversely, the reinforced core rock exhibits a higher strength under a



(a) LDP curves of pre-extrusion deformation



(b) Displacement release coefficients at the tunnel face section

Fig. 13. Surrounding rock deformations under different GSIs.

higher σ_3 or I_1 value, and a greater proportion of the surrounding rock is under elastic stress conditions, which provides a good restraint against the pre-convergent deformation of the unexcavated surrounding rock and so enhances its stability.

The ultimate purpose of reinforcement is to reduce the 3D spatial effects close to the tunnel face no matter what methods are applied (i.e., the New Austrian method or the ADECO-RS method). It can be seen that different reinforcement parameters (i.e., GSIs) produce similar curve shapes based on the unified relationship of the pre-extrusion and pre-convergence deformations, as shown in Fig. 14(e)–14(g). We could therefore conclude that the pre-convergent deformation of the surrounding rock could be successfully controlled as long as the extrusion deformation was controlled. This uniformity seems independent of the

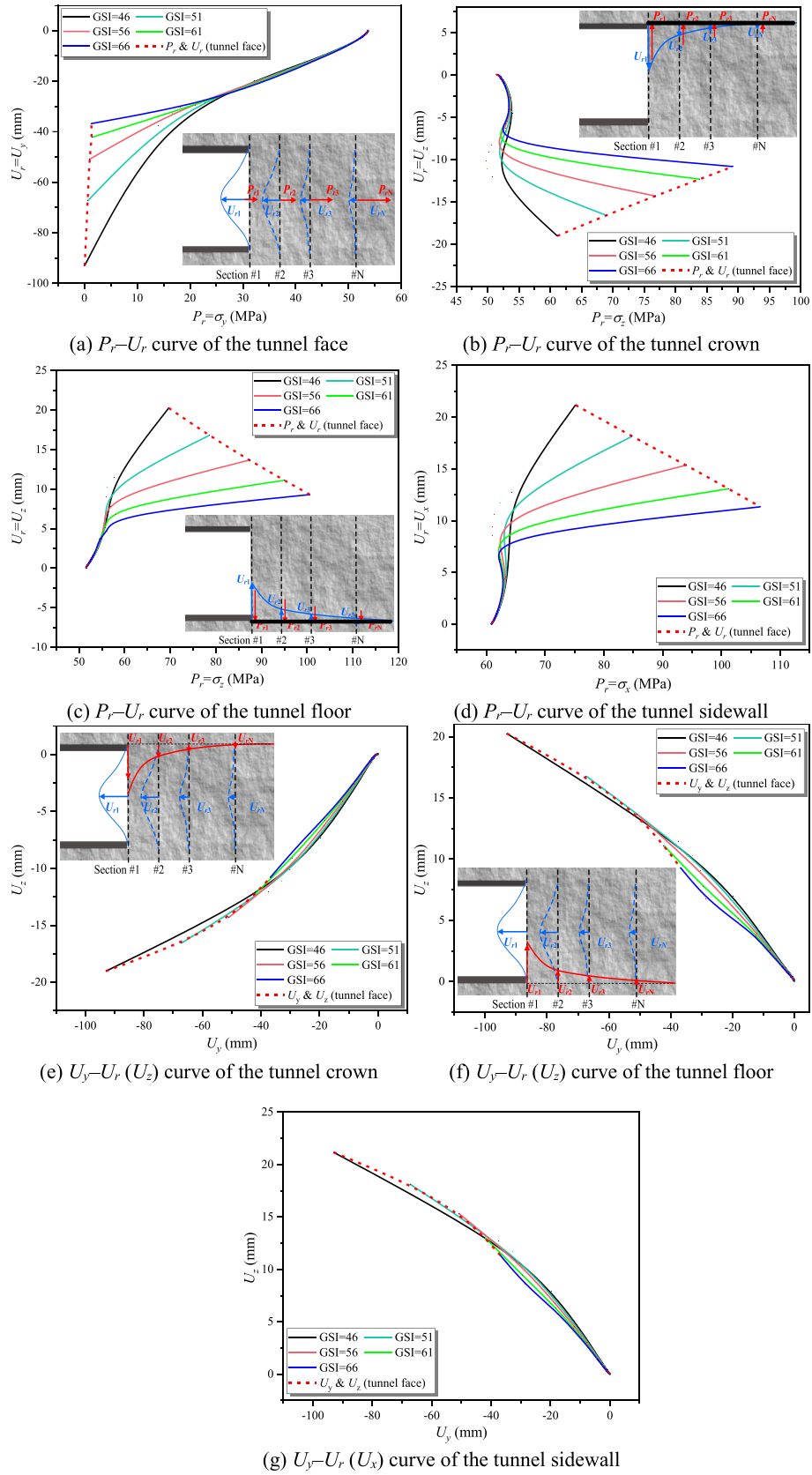


Fig. 14. Convergence-constraint curves of surrounding rock.

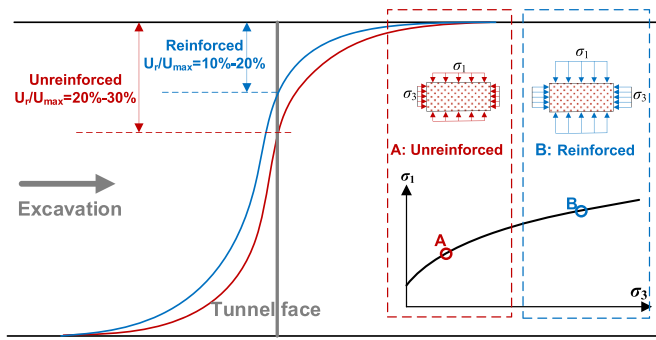


Fig. 15. Mechanical mechanism of the active control of advanced core rock mass.

control method employed or the reinforcement parameters (GSI_s) adopted. In addition, it further confirms the interaction mechanism and close linkage relationship among the convergence, pre-convergence, and pre-extrusion deformations of the surrounding rock and the tunnel face. Furthermore, based on the aforementioned mechanical mechanism of active control for the advanced core rock mass and the evolution law of the instability region ($YAI_{GZZ} = 2.0$) of the surrounding rock discussed in Section 4.3, it could be concluded that the extrusion deformation of the tunnel face is the first step in deformation of the surrounding rock, and that the strength and deformation of the advanced core rock mass are key factors in determining the overall stability of the tunnel rock.

As clarified in Section 4, tensile stress may appear at the tunnel face due to deep tunnel excavation in unreinforced core rock masses. Consequently, tension damage may occur even if the tensile stress is low, since the tensile strength of the rock mass is almost zero, and it must be controlled in a compressive state when playing a supporting role due to its excellent compression performance. The strength of the rock under a low confining pressure is particularly sensitive to changes in the confining pressure based on the nonlinear relationship between the rock mass strength and the confining pressure. Therefore, the method of controlling the deformation of the core rock mass to control the unloading stresses has a significant enhancement effect on the rock mass strength, especially for the rock mass close to the tunnel face, which can exhibit a low compression or tensile stress.

The New Austrian or ADECO-RS methods are the two main techniques used for tunnel excavation and stability control. The New Austrian Method emphasizes the restraining effect of the reserved core rock on the deformation of the tunnel surface, while the ADECO-RS method improves the rigidity and strength of the core rock using pre-reinforcement. The strength reduction of the rock mass caused by excavation unloading and disturbance is restrained by suppressing the extrusion deformation of the tunnel face. In this case, the self-supporting effect and strength potential of the rock material are fully utilized to achieve the purpose of stabilizing the surrounding rock. Consequently, both construction methods achieve active control of the tunnel rock. By controlling the deformation of the core rock mass, greater amounts of stress are retained inside the rock, thereby increasing the strength of the rock mass and reducing the redistribution of stress caused by excavation unloading. Deep rock tunnels are often under strong disturbance, wherein the unloading effect produced by excavation becomes more obvious. An active control method could therefore be employed to give full play to the strength potential of the rock, wherein the rock material is used as a natural support structure. This kind of active control design will play an important role in the design and construction of deep and ultra-deep tunnels.

7. Conclusions

To overcome the limitations of conventional indoor and back

analysis-based parameter acquisition methods, we developed a digital (intelligent) acquisition technique to quickly, automatically, and accurately obtain the in-situ strength parameters of rock mass, e.g. the GSI and D parameters, based on in situ three-dimensional (3D) photography and laser scanning. This technique showed promise to achieve 3D forward and dynamic analysis for rock tunnel construction when the suggested digital technique and GZZ-based 3D numerical modeling were considered, and it was successfully applied to simulate a deep tunnel excavation.

In addition, slight differences were observed for the deformations when different strength criteria were applied in a shallow tunnel, while significant differences were found for a deep tunnel, where the plastic zone is widely distributed. The deformations of the surrounding rock based on the 2D criteria (e.g., the HB and MC criteria) were much larger than that observed using the GZZ criterion in a deep tunnel, especially in terms of the extrusion deformation of the tunnel face; this result indicates the significant influence of σ_2 on the rock mass strength and deformation.

Furthermore, the GZZ-based numerical results were in better agreement with the field monitoring deformation data for shallow, medium, and deep buried tunnels compared with the results obtained using the HB and MC criteria, thereby verifying the reliability and applicability of the GZZ strength criterion. Moreover, we found that the unloading effect and the tunnel face extrusion deformation of the deep tunnel caused by excavation were more obvious than those of the shallow tunnel. The instability of the tunnel face was the most important factor in determining the overall instability of the surrounding rock in deep tunnel, wherein the tunnel face became unstable before the surrounding rock. The reinforcement of the core rock mass could reduce the stress unloading effect and increase the overall stress level inside the core rock mass, thereby promoting transition of the core rock from the plastic to the elastic stress state, wherein the σ_3 value of the tunnel face rock changes from a tension to a compressive condition, thereby enhancing the face stability.

We also found that the curves of the pre-extrusion vs pre-convergence deformations under different GSI_s were similar, and so the uniformity law suggested that pre-convergent deformation of the core surrounding rock could be successfully controlled as long as the extrusion deformation was controlled. This observation further confirmed the interaction mechanism and close linkage relationship among the convergence, pre-convergence, and pre-extrusion deformations of the surrounding rock and the tunnel face. The design experience and theory of shallow tunnels, which emphasize the plane strain assumption without considering the 3D tunnel face effects, should not be directly adopted for deep tunnels. We consider that the stress and deformation control of the core rock mass can be treated as an active control method to render the rock itself a natural support material to give full play to its strength potential, and so may therefore make a valuable contribution towards the construction of deep and ultra-deep rock tunnels.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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